

# 3. Ohm's Law

The first lesson with numbers

Voltage, current, resistance — and a meter in your hand

Lesson 3 · about 90 minutes · everyone builds; battery power only

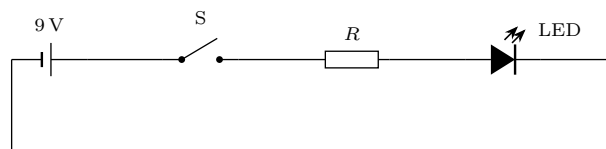
## What you need

- A digital multimeter (each, if possible)
- Solderless breadboard and jumper wires
- 9 V battery and clip; two AA cells and a holder
- Resistors: 100  $\Omega$ , 220  $\Omega$ , 470  $\Omega$ , 1 k $\Omega$ , 10 k $\Omega$
- Several LEDs
- A small torch bulb and holder
- A toggle or push switch

## Predict first

An LED is connected to a 9 V battery through a 470  $\Omega$  resistor. You then swap it for a 220  $\Omega$  resistor. Does the LED get brighter, dimmer, or stay the same — and roughly by how much? Write a number, not a word.

## Do it



current is the same everywhere in one loop

1. **Build the loop.** Battery, switch, resistor, LED, back to the battery. The LED only works one way round — if nothing lights, turn it around before you assume anything is broken.
2. **Measure voltage.** Voltage is measured *across* a component, so the meter goes in parallel: red probe one side, black the other, without disturbing the circuit. Measure across the battery, across the resistor, and across the LED. Add the last two together and compare with the first.
3. **Measure current.** Current is measured *through* the circuit, so the meter must become part of the loop: break the circuit and put the meter in the gap. Record it in milliamps.
4. **Swap resistors.** Repeat for 100, 220, 470, 1 k and 10 k $\Omega$ . Fill in the table:  $R$ , measured  $V$  across the resistor, and measured  $I$ . Then compute  $V/I$  for each row.
5. **Series and parallel.** Two 470  $\Omega$  resistors in a row, then side by side. Measure the total resistance of each arrangement before you reason about it.

## What it means

|                                              |                                                                                                                                                                                                                                                     |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>The three quantities</b>                  | <b>Voltage</b> is the push, measured between two points. <b>Current</b> is the flow, measured through one point. <b>Resistance</b> is how hard the path fights the flow.                                                                            |
| <b>Ohm's law</b>                             | $V = IR$ . Your $V/I$ column should have come out close to the marked resistance every time. You did not assume the law — you measured it into existence, which is the correct order.                                                               |
| <b>Voltages add around a loop</b>            | Your resistor and LED voltages summed to the battery voltage. Every volt the battery supplies is spent somewhere before the current gets back.                                                                                                      |
| <b>Current is the same all the way round</b> | A series loop has one current. Nothing is used up; charge that leaves comes back.                                                                                                                                                                   |
| <b>Why the LED needs the resistor</b>        | An LED is not a resistor — above its turn-on voltage its current rises almost without limit. The resistor sets the current. Without it the LED passes as much as the battery can give and destroys itself. Try it once, deliberately, with a spare. |
| <b>Series and parallel</b>                   | In series, resistances add. In parallel the total is <i>smaller</i> than either — two paths carry more than one.                                                                                                                                    |

## Habits to build now

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|                            |                                                                                                                                                                                  |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>One hand</b>            | Start now, at 9 V where it does not matter. Meter probe in one hand, other hand off the bench. By Lesson 11 you will not have to think about it.                                 |
| <b>Meter mode first</b>    | Look at the dial before every measurement. A meter left in current mode and then put across a battery is a short circuit through a fuse, and on a big supply it is an explosion. |
| <b>Power off to rewire</b> | Disconnect the battery before changing the circuit, every time, even though nothing here can hurt you.                                                                           |

### The link to the coil

Everything you will build later is measured with this meter and reasoned about with this law. The bleeder resistors that drain the capacitor bank, the current limit of the transformer, the safety of a discharge — all of it is  $V = IR$  with big numbers. 10 mA through a person is roughly what kills them; work out for yourself what resistance across 9 kV that implies, and whether dry skin will save you.

**This lesson is low voltage.** Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.

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